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Review

Blockchain-Assisted Mission Trust in Air–Surface–Underwater Unmanned Systems: A Review and Conceptual Framework

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Abstract

Air–surface–underwater missions combine heterogeneous vehicles, intermittent links, and administrative domains. A reliable decision must connect physical observations, protected records, shared authority, and fitness for use. This structured narrative review applies the Object–Assurance–Lifecycle (OAL) framework to 39 full-text mechanism studies through 89 scoped claim records. Comparative synthesis distinguishes sender authentication from observation validity, platform integrity from task suitability, and replicated-state recovery from mission-data invalidation. Domain-stratified and sensitivity analyses expose the effects of inclusion and coding choices on the coverage map. The dependency graph identifies products affected by changed inputs or authority, and the context–temporal–graph loop organizes their reassessment. A partition–revocation–reconciliation case specifies coalition roles, a Fabric-oriented application pattern, constrained verification, and an acoustic-link payload budget. A synthetic event replay verifies signatures, inclusion proofs, and temporal and authorization rules. Under matched rules, a policy-validating witnessed log yields the same decisions. Ledger value is therefore conditional on joint-governance requirements and implementation cost. Mission fault and communication requirements guide consensus selection; proposed tests compare decision quality and recovery under matched conditions. The synthesis connects mechanism evidence to explicit composition requirements for cross-domain mission decisions.

Keywords: mission trust; unmanned systems; UAV–USV–UUV cooperation; blockchain; data provenance; remote attestation; zero trust; maritime autonomy

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1. Introduction

Air–surface–underwater missions combine unmanned aerial vehicles (UAVs), unmanned surface vehicles (USVs), and unmanned underwater vehicles (UUVs) for observation, inspection, search, and environmental monitoring. Their complementary mobility and sensing capabilities create dependencies across radio and acoustic links, gateways, processing services, and administrative owners [1–3]. UAV and marine-system reviews identify mobility, limited energy, intermittent communication, localization uncertainty, and difficult recovery as operational constraints [4–7]. A mission decision must combine evidence generated under these different conditions.

This creates a *mission-trust gap*. Authentication establishes a principal's identity; a content hash binds bytes; replicated consensus establishes agreement under a fault model. A relying decision additionally requires evidence that the observation is physically credible, sufficiently current, and suitable for its intended consequence. A signed sonar report can be inaccurate, a genuine image can be stale, and a correctly ordered record can originate from a captured device. UAV and underwater-security research describes the corresponding spoofing, jamming, capture, routing, and data-manipulation threats [8–12]. Connecting these controls to the eventual use of a product requires explicit claim semantics.

Blockchain proposals address swarm authentication, data acquisition, membership, provenance, and audit [13–16]. Reviews of UAV and maritime applications describe coordination and sharing opportunities [17–21]. Their value depends on which parties must jointly govern the state, which faults must be tolerated, and whether an authenticated database or independently witnessed log already satisfies those requirements. The blockchain oracle literature names the external-input problem directly: agreement about recorded assertions does not establish their correspondence with physical events [22,23].

Adjacent mechanisms supply complementary evidence. Zero trust provides resource-specific authorization [24,25]; remote attestation separates device evidence, appraisal, and relying decisions [26]; provenance records producers and transformations [27,28]; and trust inference models reputation, uncertainty, and changing behavior [29–31]. These mechanisms use different vocabularies and evaluate different claims. Their composition is the central subject of this review.

The proposed *Object–Assurance–Lifecycle* (OAL) framework relates entities, platforms/devices, links/paths, and data products to four assurance layers across six lifecycle stages. A typed dependency graph and context–temporal–graph (CTG) loop preserve shared causes, event time, and uncertainty when applying this structure to a decision. OAL supplies the claim organization; established inference, provenance, policy, and consensus mechanisms supply implementations.

The review makes three connected contributions. First, it applies OAL to 39 full-text mechanism studies, comparing the evidence supplied by complementary mechanisms and testing coverage sensitivity to inclusion and coding choices. Second, a partition–revocation–reconciliation case connects the framework to a specified permissioned-ledger pattern, an executable policy replay, and a communication budget. A policy-validating witnessed-log baseline exposes where the two architectures provide equivalent decisions. Third, requirement-based consensus selection and falsifiable research propositions translate the synthesis into an evaluation agenda.

Civil missions and publicly documented coalition settings provide the application context.

2. Review Scope and Methods

2.1. Identification and Selection

This structured narrative review examines unmanned cooperation, communication and security, trust inference, provenance, attestation, and distributed governance. OpenAlex was queried on 6 August 2026 using 32 English search strings, a publication-date filter starting on 1 January 2016, and the 25 highest-ranked results per query. Deduplication by DOI or OpenAlex identifier produced 571 candidate records. Queries included "blockchain UAV swarm", "underwater wireless sensor network trust management", "UAV USV UUV cooperative systems", "remote attestation IoT survey RATS", and "blockchain network partition finality". Citation following and targeted retrieval added earlier foundations and official specifications. Crossref metadata supported bibliographic checking; it was not a second independent screening database.

Selection favored English sources that described cross-domain operating constraints, specified a security or trust mechanism applicable to a mission boundary, or supplied a relevant standard, dataset, or simulator. Duplicate records and sources without a substantive connection to those questions were excluded from the synthesis. Reviews supplied context and citation leads; original mechanism studies supplied the analytical sample. Foundational work predating 2016 was retained for reputation, distributed consistency, provenance, and feedback architectures. Targeted publisher, author-repository, and official-source retrieval checked developments through 25 October 2026. Table 1 summarizes the scope and evidence roles.

2.2. Full-Text Coding and Synthesis

The analytical sample comprises 39 original mechanism contributions with inspected full texts. Its direct-domain stratum contains 11 UAV, three maritime AIS, and five intended underwater studies. Twenty transferable studies address reputation, fog offloading, provenance, attestation, access control, shared state, or observation appraisal. Application domain and actual evaluation setting are recorded separately: a terrestrial-data experiment proposed for underwater use remains a proxy evaluation. The four generic consensus comparators form an identifiable, removable subset. These purposively selected mechanisms support comparison across assurance boundaries; selection does not estimate prevalence in the field.

The bibliography contains 123 references. The 39 coded studies are a subset; the remaining 84 support operational context, conceptual foundations, standards, related synthesis, or evaluation resources. Table 2 identifies the coded study groups and their sources. For each study, extraction recorded the mechanism claim, supporting passage, source version, and evaluation scope.

The coding unit is an explicitly described mechanism claim. Each of the 89 records associates one object with one assurance layer and the supported lifecycle stages. A paper can contribute several claims, each coded only for the object–layer–stage associations supported by the cited passage. A heatmap cell counts each study once for that exact object–layer–stage combination. Counts describe coverage, not additive sample sizes or effect sizes. Layer co-occurrence within a paper identifies a possible interface; validated transfer between claims requires its own evidence.

Evidence profiles record conceptual argument (L0), analytical/security reasoning (L1), simulation or recorded-data evaluation (L2), implemented prototype/testbed (L3), physical operating trials (L4), and sustained multi-party operation (L5). These categories are complementary evidence forms, not an ordinal quality score. A profile is attached to the stated mechanism and documented evaluation scope; an implementation benchmark does not validate every assurance claim made in its surrounding architecture. Recorded field data analyzed offline are coded L2; a hardware prototype without described operating conditions is coded L3. Ambiguous deployment claims are assigned the narrower evidenced category and explained in the extraction note.

Coding used one extraction process with passage checks; independent duplicate coding and inter-rater agreement were not performed. Sensitivity analyses test the effects of domain balance, study inclusion, ambiguous identity assignments, evidence-profile coding, application/data mismatches, and removal of individual studies (Section 5.4). Empty cells denote no coded claim.

Table 1. Scope and evidence roles in the synthesis.

Role	Included Material	Use in the Synthesis
Operational context	UAV, USV, UUV, gateways, shore services; cooperative control and communication	Mission dependencies and deployment assumptions
Analytical sample	39 full-text mechanism studies; 89 claim records; two application strata	OAL coverage, scoped L0–L5 profiles and inclusion/coding sensitivity
Contextual foundations	Reviews, reputation and distributed-system foundations, standards and governance cases	Vocabulary, alternatives and design requirements
Evaluation resources	Datasets, simulators and benchmark practices	Candidate tests; no inferred integrated mission validation
Worked case	Executable policy replay and exact application-byte accounting	Rule consistency and illustrative design trade-offs

3. Operational Context and Trust Foundations

3.1. A Heterogeneous Mission System of Systems

UAVs contribute rapid repositioning, line-of-sight radio relays, and electro-optical sensing. USVs carry larger energy stores and antennas and bridge acoustic and radio networks. UUVs operate close to underwater targets but often rely on low-rate acoustic links or delayed surfacing for large transfers. Communication surveys describe the resulting trade-offs among range, bandwidth, latency, energy, and alignment [32–35]. Topology, clock quality, and reachability can change within a mission.

Consider a civil coastal-response coalition comprising authority A, research operator B, communications provider C, and contracted data provider D. B operates a UUV and UAV; C supplies the USV relay; A approves mission authority; D contributes independent data and audit services. Each organization retains its administrative identity. A local operator can authenticate a device while lacking authority to change the coalition's membership, evidence policy, or revocation history. The case in Sections 6.4 and 7.3 uses this coalition to specify which changes require joint approval.

A USV may simultaneously relay acoustic traffic, cache data, provide time, and host fusion. Failure at that shared dependency affects several apparently independent products. End-to-end source signatures can preserve origin integrity through an untrusted relay, while selective withholding and common clock or transformation errors remain relevant. Figure 1 separates the constrained evidence path from shore-based shared-state services.

Raw imagery, sonar, detailed telemetry, and attestation evidence remain encrypted off-chain. Shared records contain commitments, membership, policy/model versions, revocations, and audit judgments. Delay-tolerant forwarding can use Bundle Protocol version 7 and BPSec integrity/confidentiality services [36,37]. Store-and-forward delivery preserves messages across contacts; authority expiry and evidence-age policy are separate application rules.

3.2. Mission Trust and the Relying Decision

Mission trust is a context-specific decision to rely on an object for a stated purpose, given available evidence and the consequences of error. *Trustworthiness* denotes the supporting properties: identity, measured endpoint state, link availability, provenance, freshness,

observation quality, and accountable governance. The same image may support broad situational awareness while being unsuitable for collision avoidance.

For object o, time t, context c, and mission use m, appraisal and decision have distinct interfaces:

$$\begin{aligned} T_o(t,c,m) &= \mathcal{A}(o,\mathcal{E}_{\leq t},c,m;\theta), \\ D_{r,o}(t,c,m) &= \pi_r(T_o(t,c,m),\mathcal{C}_t,m). \end{aligned} \tag{1}$$

Here $\mathcal{E}_{\leq t}$ contains evidence received by decision time, including observation times and lineage. The appraisal $\mathcal{A}$ produces claim-specific outcomes and uncertainty under assumptions θ. Relying party r applies policy π_r under authority, deadline, consequence, and resource constraints $\mathcal{C}_t$. The result may permit, restrict, defer, or reject use. Subjective opinions, intervals, ordinal states, and reputation values can implement this interface [31,38,39]. A probabilistic interpretation requires a defined event and calibration evidence.

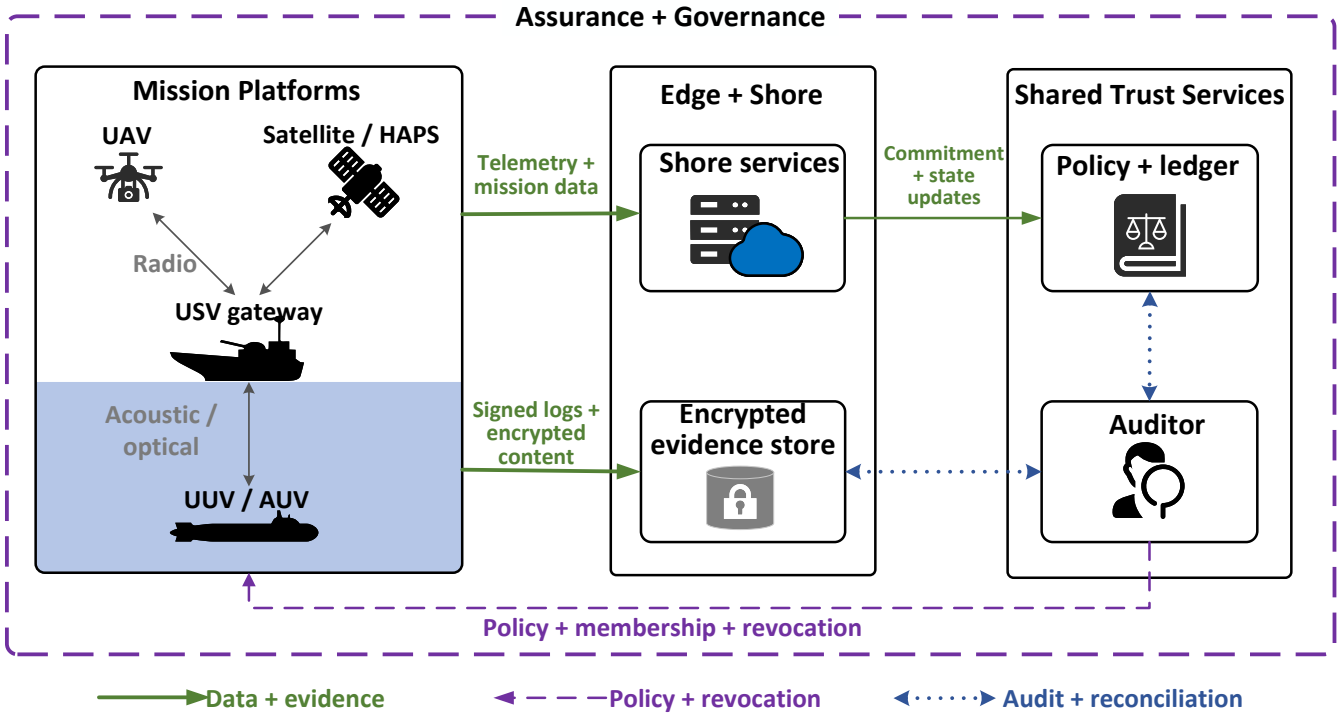


Figure 1. Cross-domain reference architecture. Shore organizations host peers and ordering nodes; the USV is a forwarding/cache dependency. Local sensing and safety continue within delegated authority during a gateway partition.

A *trusted data environment* is the socio-technical setting that produces, transforms, shares, and governs this evidence. It contains principals, devices, paths, artifacts, and accountability processes. Zero trust contributes resource-centered policy and enforcement [24]; OAL records the additional observation, lineage, timing, and consequence claims relevant to mission reliance.

3.3. Four Complementary Assurance Claims

Observation validity concerns correspondence with the physical world. Commitment and provenance integrity concerns protected bytes and recorded origins or transformations. Ledger and governance consistency concerns authorized shared-state transitions. Mission-use suitability concerns the particular decision and consequence. Figure 2 shows the distinct questions and supporting evidence.

A forged observation signed with a stolen key can pass signature verification. RATS appraisals support scoped measured-state claims [26], while physical diagnostics and corroboration support observation claims. The relying policy composes the applicable results, with explicit responses to failed or unknown requirements.

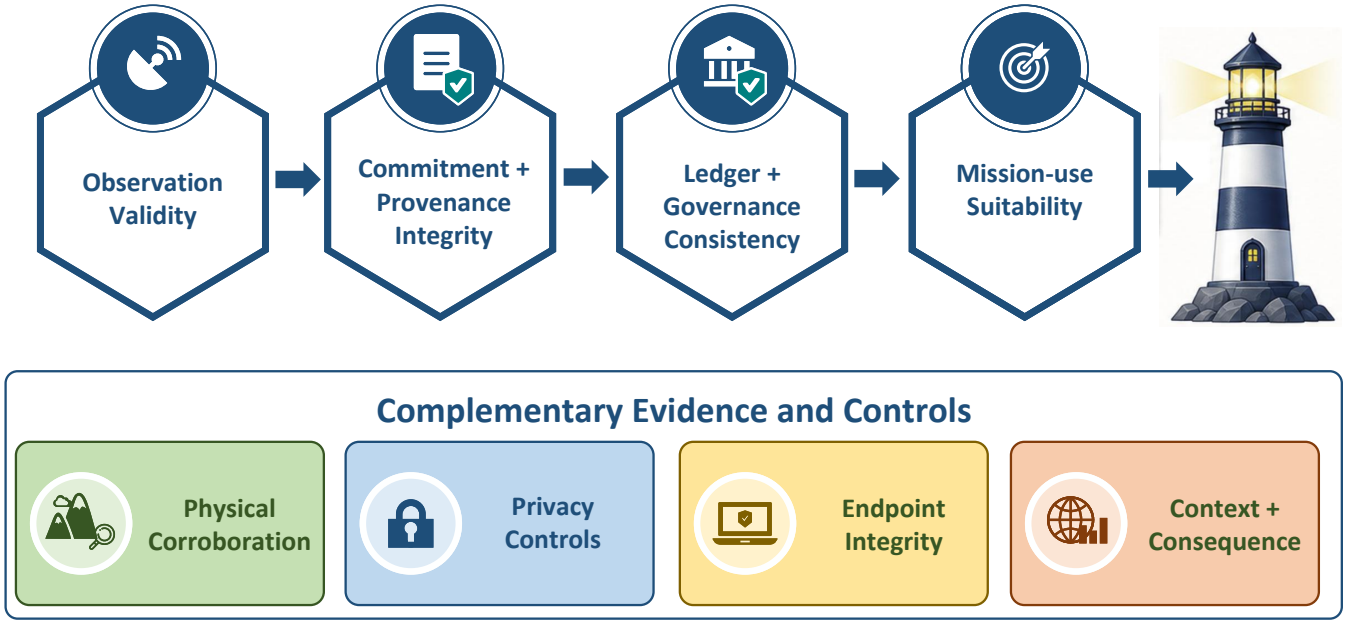


Figure 2. Four assurance claims and their evidence. The relation is an argument structure; an implementation may evaluate the checks in a different order.

4. Complementary Research Landscapes

Cooperative navigation and formation studies establish interactions among heterogeneous vehicles [40–43]. Their use of *consensus* concerns coordination of physical states. Ledger consensus concerns agreement on a replicated record under a fault model. OAL relates their outputs through identified observations, transformations, authorities, and intended uses.

4.1. Security and Evidence Appraisal

Security surveys organize spoofing, jamming, capture, malicious routing, malware, and false-data injection by target and attack mechanism [8,11,44,45]. Underwater fading, noise, Doppler, and topology change can also cause loss or disagreement [33,46,47]. A trust estimate therefore needs a named property and an uncertainty status: an anomaly may justify restricted use while its cause remains unresolved.

Decision-tree and hidden-Markov approaches illustrate underwater trust inference [48,49]. The former evaluates simulated network behavior with an initial attack-free training period; the latter includes a numerical illustration with ten generated observations. These studies support trust inference within their specified network and observation models. Reputation-based peer selection [39] similarly supplies an appraisal mechanism whose meaning depends on the service and observed behavior. Survey taxonomies distinguish direct observations, recommendations, context, and attacks on reputation [29,30,50,51].

Observation appraisal supplies a complementary strand. Signal-feature classification, network-assisted detection, and recorded-data validation have been studied for GNSS spoofing [52–54]. Multi-view opinion aggregation combines uncertain observations under

declared assumptions [55]. These methods address evidence about the physical world. Their outputs become inputs to a relying policy with a specified consequence and validity interval.

4.2. Provenance, Attestation, and Access

W3C PROV models entities, activities, and agents [27,28]. An OAL entity is an acting principal and usually maps to a PROV agent; an OAL data product maps to a PROV entity. Decision provenance extends the analysis along the data flows that lead to organizational or automated decisions [56]. It supports accountability when event capture and retained records cover the relevant transformations.

Attestation supplies measured endpoint-state evidence. Architectural proposals discuss attestation of ledger participants and autonomous IoT security [57,58]; implemented work evaluates attested channels and dual attestation of embedded devices and deployed ML artifacts [59,60]. RATS separates evidence, verifier appraisal, and the relying decision [26]. Zero trust and device baselines provide authorization and endpoint-capability requirements [24,25,61]. These boundaries matter during disconnection, when cached results and delegated permissions age independently.

4.3. Shared Records and Governance

The coded UAV studies combine authenticated identities or data acquisition with ledger state [13–16]. Fabric provides an implemented execute–order–validate architecture [62]; IoT platforms apply it to sensing-record integrity, off-chain provenance, and access policy [63–65]. Food, health, and digital-twin studies extend these patterns to forecasting, federated learning, and lifecycle coordination [66–68]. Their recorded mechanisms and evaluation settings are summarized in Table 2 and mapped in Section 5.

Table 2. The 39 coded studies and their principal evidence scope. Profiles apply to scoped claims and may overlap within a study.

Identifiers	Mechanism Focus	Profiles	Evaluation Scope
S01–S04	UAV identity/authentication/acquisition [13–16]	L1, L2, L3	Security analysis, simulation and implemented components; BUS includes UAV and IoT hardware
S05–S07	Underwater trust and P2P reputation [39,48,49]	L1/L2	Network or numerical evaluation and peer-selection simulation
S08–S10	Fabric, sensing records, Hadoop provenance [62–64]	L3	Server and embedded-device prototypes; record processing and retrieval
S11–S14	Attestation and autonomous IoT security [57–60]	L0 or L3	Two architectural contributions; two implemented attestation studies
S15–S17	Food/health provenance and access control [65–67]	L2 and/or L3	Forecasting, FL applications and controlled access-policy requests
S18–S19	Decision provenance and digital twins [56,68]	L0	Conceptual data-flow and lifecycle architectures
S20–S23	GNSS and multi-view observation appraisal [52–55]	L1/L2	Signal/learning experiments, recorded data, and aggregation analysis
S24–S26	Visual corroboration, UAV reports, fog offloading [69–71]	L0/L2	Recorded trajectories, agent simulation, and modeled partner/resource selection
S27–S29	Maritime AIS provenance and message protection [72–74]	L3	Receiver/ledger benchmark and two software protection demonstrations
S30–S31, S39	UAV reputation, governance, and routing [75–77]	L0/L2/L3/L4	Simulated trust updates; S31 includes a physical UAV-node experiment and a separate container testbed
S32–S34	Underwater signcryption, routing, and FL [78–80]	L1/L2/L3	Desktop cryptography, simulated routing and terrestrial-data proxy learning
S35–S38	Generic consensus comparators [81–84]	L1/L3	Protocol reasoning and server/cloud prototypes; fault models and recovery paths differ

The oracle problem remains relevant across these applications: a ledger preserves the accepted assertion and its provenance, while external validity depends on the data source and appraisal [22,23]. Transparency logs offer a natural alternative. Certificate Transparency specifies inclusion and consistency proofs, with external monitoring or witnessing needed to address inconsistent views [85,86]. Sigstore combines signing with transparency for software artifacts [87]. A witnessed log can also carry jointly authorized state transitions. The distinction to evaluate is therefore the required validation and governance service, together with its implementation cost, rather than the presence of hash chaining alone.

5. The OAL Framework and Literature Mapping

OAL classifies *what is judged*, *which claim is sought*, and *when that claim is needed*. A mechanism can support several cells; some combinations are inapplicable to a mission. Dependencies between claims are represented separately so that classification does not imply evidential independence.

5.1. Objects, Assurance Layers, and Lifecycle

An *entity* is an accountable principal: an organization, operator, service, or autonomous agent with delegated authority. A *platform/device* is the endpoint executing a function, including vehicles, gateways, sensors, and hosted compute instances. A vehicle can be considered an entity when exercising delegated authority and a platform when its firmware is measured. A *link/path* comprises physical media, intermediate nodes, security associations, and changing quality along an end-to-end route. A *data product* is an identified version of an observation, track, map, model, command, policy, or audit record. Corrections and transformations create new version identifiers. Table 3 summarizes the object-specific evidence and relying questions.

Table 3. OAL objects and claim-specific evidence.

Object	Evidence	Relying Question
Entity (E)	Credential, role, delegation, accountable actions	Who acts and with which authority?
Platform (P)	Boot/runtime measurements, configuration, self-test, energy	Which endpoint executes, and in what measured state?
Link/path (L)	Authentication, route, delay, loss, custody, and medium state	What protection and timeliness held along the path?
Data product (D)	Signature, content ID, times, lineage, uncertainty, residuals	Which version is this, and what supports its intended use?

The assurance axis uses O (observation validity), C (commitment and provenance integrity), G (ledger and governance consistency), and M (mission-use suitability). O uses physical bounds, calibration, diagnostics, and corroboration. C binds producer, version, and transformation assertions through signatures and provenance. G addresses jointly authorized membership, order, policy, and recovery state. M applies purpose, freshness, consequence, uncertainty, and authority requirements. When a shared-state service is not needed, ledger-specific claims are inapplicable; equivalent governance claims may be met by another service.

Policies, models, and reference values are versioned data products; their deployed instances reside on platforms, and their approval belongs to entities. This treatment makes the assurance infrastructure subject to the same analysis as mission observations. A cell specification identifies the object/version and required property, then gives the supporting evidence and validity interval. It also names the acceptance rule, responsible party, and response to missing evidence. Table 4 summarizes the lifecycle stages and state transitions.

Table 4. Lifecycle stages and relevant state transitions.

Stage	Required Assurance	Examples of Retained State
Admission (A)	Authorized principal and acceptable endpoint	Role, delegation, attestation reference
Sensing (S)	Credible time, place, and observation	Product ID, quality and acquisition commitment
Transmission (T)	Protected content and adequate path	Relay/custody event and batch root
Fusion/sharing (F)	Identified inputs, transformation and access	Lineage, model and policy version
Decision/action (D)	Fit for the specified consequence now	Decision rationale and responsible principal
Recovery/audit (R)	Scope of invalidation and restored authority	Revocation, dispute, reconciled history

5.2. Observed Coverage in the Coded Sample

Figure 3 maps 89 claims from 39 studies. Twenty-five of the 96 possible cells contain a coded claim. Data governance during processing/fusion (D/G/F) is the largest cell, with 16 studies. The 71 zero cells have no coded claim in this sample; their applicability depends on the mission. Cell counts also overlap: one study can protect a record, replicate it, and support its later audit.

The strata expose different emphases. Among 19 studies targeting UAV, maritime AIS, or underwater applications, entity governance at admission (E/G/A) is the largest cell, with six studies. Among 20 transferable studies, D/G/F leads with 11. Thus, the overall concentration partly reflects the inclusion of provenance and shared-state infrastructure. Each claim was checked against its supporting passage and evaluation setting.

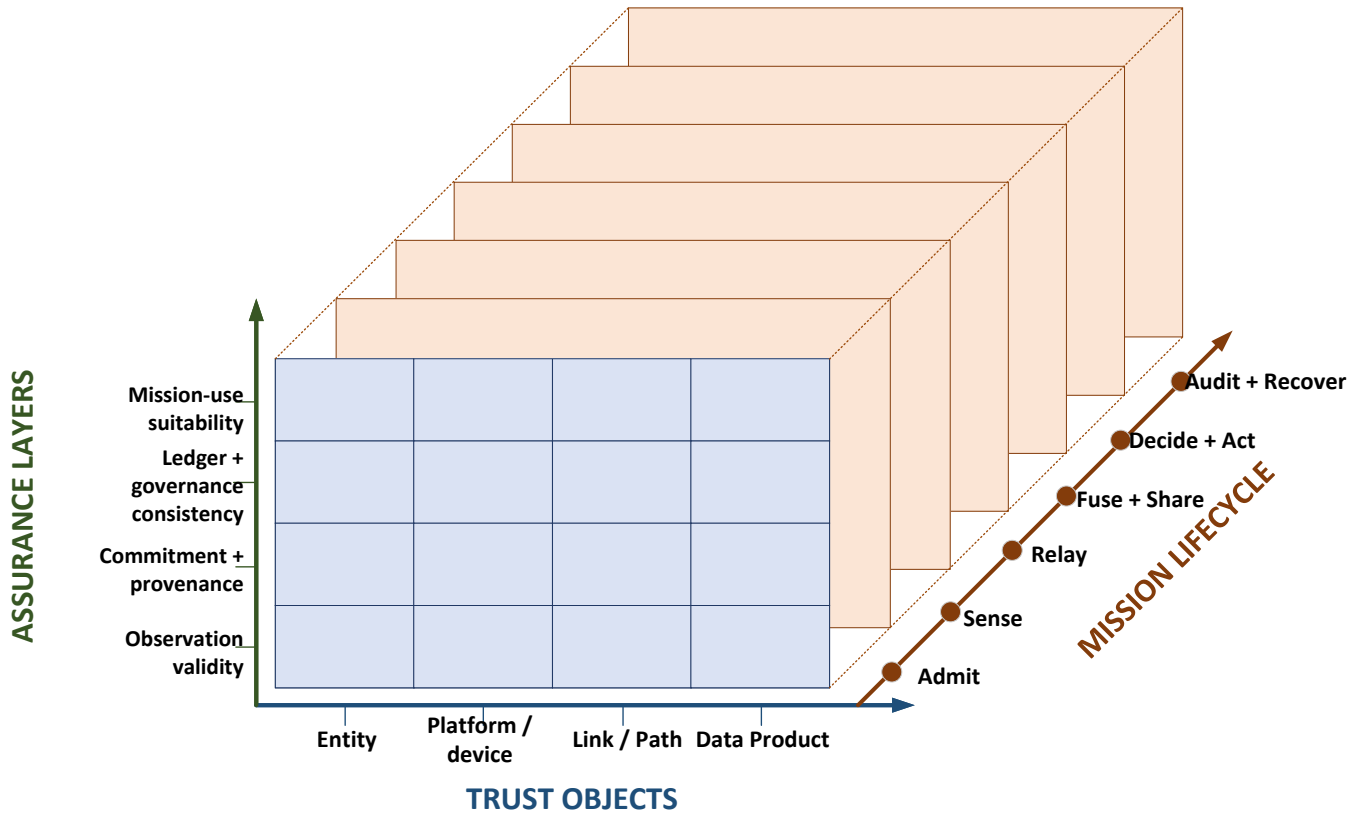
5.3. What the Comparison Changes in a Design Review

The OAL axes separate the evidence supplied by a mechanism from the requirements of its next consumer. Table 5 compares mechanisms that share an application label or trust vocabulary but support different relying decisions. Table 6 reports sensitivity to domain and coding choices.

Table 5. Claim boundaries exposed by the OAL comparison. The final column states the additional requirements for composing these mechanisms.

Mechanism Comparison	Evidence Supplied	Consequence for Composition
AIS protection and receiver ledger (S27–S29)	Sender/message protection (E/C/T, D/C/T) and receiver-submitted history (D/C and D/G at F/R) have different trust boundaries	Preserve the original sender binding; require observation appraisal and freshness before using a position for action
Visual and multi-observer corroboration (S24–S25, S39)	Trajectory consistency or report agreement supports D/O at sensing/fusion under texture, geometry, and observer assumptions	Retain those assumptions with the result; authenticated agreement alone does not exclude correlated error
Attestation and partner selection (S12, S14, S26, S34)	Measured endpoint/model integrity differs from task-specific offloading or update suitability	Check both execution state and task evidence; neither credential nor trust score substitutes for the other's claim
UAV trust and routing (S30–S31, S39)	Changing behavior can trigger exclusion or path adaptation (E/M, L/M and E/G)	Reuse these continuity mechanisms, then specify authority and dependent-product handling across disconnection
Replicated-state protocols (S35–S38)	Similar D/G/F or D/G/R cells retain different fault models, synchrony and exercised recovery paths	Select assumptions before comparing latency; replicated-log recovery alone does not invalidate a derived mission product

(a) Object-Assurance-Lifecycle framework



(b) Coverage of 39 full-text studies / 89 claim records

		Admission	Sensing	Relay	Fusion	Decision	Recovery
Entity	O	0	0	0	0	0	0
	C	6	0	4	0	0	0
	G	10	0	0	0	0	3
	M	0	0	5	4	1	1
Platform / device	O	0	0	0	0	0	0
	C	7	0	0	1	0	1
	G	0	0	0	0	0	0
	M	0	0	0	1	0	1
Link / path	O	0	0	0	0	0	0
	C	0	0	0	0	0	0
	G	0	0	0	0	0	0
	M	0	0	3	0	0	0
Data product	O	0	7	0	5	0	0
	C	0	4	6	11	1	8
	G	0	0	0	16	0	11
	M	0	0	0	1	4	0

O: observation C: commitment/provenance G: governance M: mission use

Evidence profiles (overlap): L0 6 L1 11 L2 20 L3 20 L4 1 L5 0

Cells count distinct studies. Zero means no coded claim in this sample.

Figure 3. OAL framework and mapping of 39 full-text mechanism studies. (**a**) The object, assurance, and lifecycle dimensions. (**b**) Distinct studies per exact cell, computed from 89 scoped claim records. Evidence-profile counts overlap. Zero denotes no coded claim, without establishing applicability or global research absence. Table 6 separates domain and inclusion effects.

Table 6. Sensitivity of the coverage map. Occupied cells are descriptive counts, not a completeness score. D/G/F denotes data governance at fusion; E/G/A denotes entity governance at admission.

Included Studies / Coding Variant	Studies	Cells	Largest Cell (Count)
All studies	39	25	D/G/F (16)
Direct-domain stratum	19	18	E/G/A (6)
Transferable stratum	20	20	D/G/F (11)
Without four generic consensus comparators	35	25	D/G/F (12)
Without conceptual-only studies	35	22	D/G/F (15)
Direct domain without proxy-data study	18	17	E/G/A (6)
Admission identities assigned only to entities	39	25	D/G/F (16)
Leave one study out (39 runs)	38	23–25	D/G/F (15–16)

The AIS comparison illustrates an assurance handover. The maritime monitoring prototype records observations submitted by authorized receivers [72]; pAIS and SecAIS instead examine protection of AIS messages [73,74]. The pAIS demonstration uses weak demonstration parameters, while SecAIS implements selected modes with simulated AIS devices. Their L3 profiles identify software implementation, not equivalent security strength or on-air validation. A relying navigation service still needs evidence about the reported position. Visual-odometry cross-checking supplies one such mechanism, with recorded-data evaluation and limitations over low-texture water or at night [69]. Multi-UAV corroboration supplies another under assumptions about observers and malicious participants [70,77]. OAL exposes where a signed, consistently stored but erroneous position could otherwise pass an underspecified acceptance rule.

UAV trust and routing studies provide precedents for continuity. Hierarchical drone trust and blockchain-assisted routing update reputation and restrict cooperation as behavior changes [75,76]. Yang et al. report a bounded experiment with eight physical UAV nodes, timed malicious forwarding, and a link interruption. This supports cooperation/path adaptation (L3/L4); their larger consensus benchmark uses a separate container environment. These mechanisms already connect observations of behavior to continued use. The additional coalition question concerns how a partitioned participant interprets changed authority and propagates invalidation to dependent products, as developed in Section 6.4.

Task suitability is equally specific. COMITMENT combines partner history with workload/capacity criteria for fog offloading [71]. Underwater signcryption evaluates cryptographic operations on a desktop, while trust/energy routing uses simulation [78,79]. TAFL-UWSN filters federated participants and updates, but its detailed evaluation uses terrestrial UNSW-NB15 data [80]. These results support distinct commitments or processing decisions; transferring them to an acoustic mission additionally requires channel, energy and data-distribution evidence.

Finally, a common cell does not imply interchangeable mechanisms. Raft and the Byzantine protocols differ in fault assumptions [81–84]. The original PBFT prototype did not implement view changes or retransmissions, so its normal-operation claim receives L1/L3, and recovery receives L1. This claim-level distinction prevents a prototype label from spreading to an untested path. Across the sample, no extracted claim establishes an integrated air–surface–underwater mission with continuity of authority, observation appraisal, and dependent-product recovery all validated.

5.4. Sensitivity to Selection and Coding

Table 6 summarizes how inclusion and coding choices change the coverage map. Removing the four generic consensus papers leaves all 25 occupied cells and the same leading cell, now with 12 studies. Removing the four conceptual-only studies reduces

occupancy to 22. Thus, infrastructure comparators increase counts, while some coverage depends on conceptual mechanisms. The direct-domain pattern remains centered on admission governance when the terrestrial-data proxy study is removed.

Reassigning the S01–S03 platform-identity rows to entity identity changes individual cell counts without changing occupancy or the leading cell. Every leave-one-out map retains D/G/F as its largest cell, with 23–25 occupied cells overall. The L4 count depends on S31: coding its bounded physical-node experiment conservatively as a testbed removes L4 without changing the heatmap. The leading cell is stable to individual-study removal, while domain stratification and claim-level comparisons expose distinct composition requirements.

6. Evidence Dependencies, Threat Propagation, and Dynamic Reasoning

6.1. A Typed Object-Dependency Graph

Let $G_t = (V_t, E_t)$ be a time-indexed directed multigraph over OAL objects and their versions or state intervals. Relations name their semantics: an entity *controls* a platform; a platform *hosts* a model or *produces* an observation; a path *transmits* a product; an entity *endorses* or *consumes* a product; and a record *commits to* another product. A transformation links its inputs, output, executing platform, responsible entity, code/model version, and event interval. In PROV, this transformation can be serialized as an activity [27]. Figure 4 shows these dependencies and their evidence states.

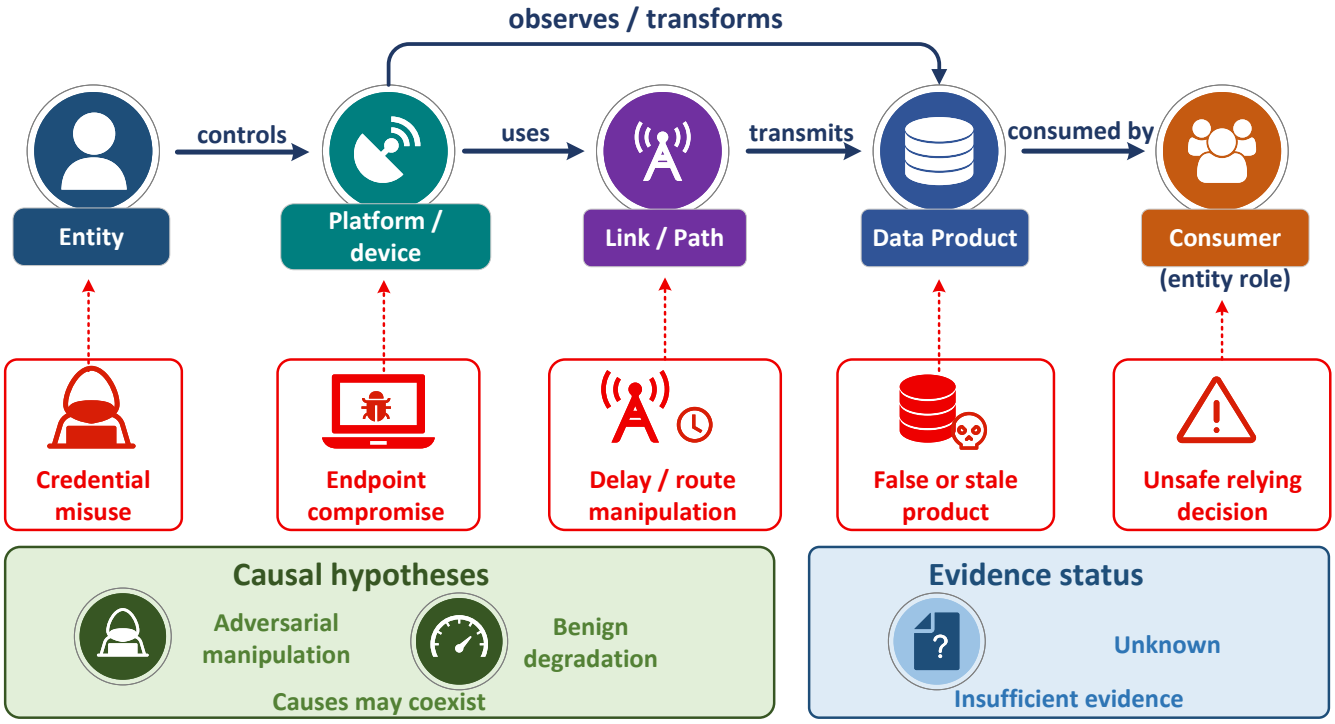


Figure 4. Typed dependencies and evidence states. Operational relations identify claims affected by a changed assumption. Benign degradation and adversarial intervention are causal hypotheses; unknown is an epistemic outcome.

Traversal identifies claims for reassessment when a key, endpoint, model, or input becomes disputed. A graph edge alone does not quantify causal influence or justify a probability update. Corroboration must account for shared sources, clocks, transformations, and administrative roots. Two products derived from one observation do not supply two

independent observations. Conversely, a shared relay need not invalidate end-to-end signatures from otherwise independent sources.

6.2. Threats, Degradation, and Missing Evidence

An analysis statement identifies the attacker or environmental cause and relevant capability, then traces the affected object and propagation relation to a mission consequence. It also records controls and residual uncertainty. For example, a compromised USV can withhold UUV batches, delaying the consumer's evidence and audit; independent contacts and protected local counters may expose gaps after reconnection. Acoustic fading can create the same delivery symptom with a different cause. The response can restrict freshness-sensitive use while attribution remains unknown. Table 7 summarizes selected cross-object failure paths.

Table 7. Cross-object analysis of selected failure paths.

Initial Condition	Affected Dependency	Evidence and Response
Stolen credential	Entity authority to signed product	Effective-time revocation; review signed descendants
Compromised endpoint	Device state to observation or transformation	Attestation scope, diagnostics, corroboration; quarantine affected outputs
Fading or jamming	Path availability to evidence freshness	Channel/context evidence; restrict stale use and retain uncertain cause
Correlated sources	Shared input to fused conclusion	Lineage and common-cause model; reduce independent-evidence count
Unavailable governance quorum	Shared policy to fresh global authority	Retain prior checkpoint; enforce bounded local delegation

6.3. The CTG Reasoning Loop

CTG uses context, temporal state, and dependencies to guide appraisal and policy action through the eight-step loop in Figure 5. Evidence capture includes binding verification; propagation identifies affected descendants for reassessment under an explicit inference rule.

MAPE-K provides monitoring, analysis, planning, execution, and shared knowledge [88,89]. CTG applies these functions to claim appraisal, separates event time from receipt time, and triggers reassessment through recorded dependencies. The JDL fusion model distinguishes object, situation, impact, and process refinement [90]. For these estimates, CTG records the identity, provenance, governance, and temporal evidence needed for a relying decision.

Implementations can use Bayesian models, subjective logic, hidden-state inference, rules, or learned classifiers. Their outputs retain the named property, context, evidence lineage, uncertainty, and permitted use. Re-estimating trust after a change requires a specified update rule, rather than an arbitrary decrement along graph edges. A hard authorization condition can coexist with a graded quality estimate; the choice to make one condition non-compensable belongs to the mission policy.

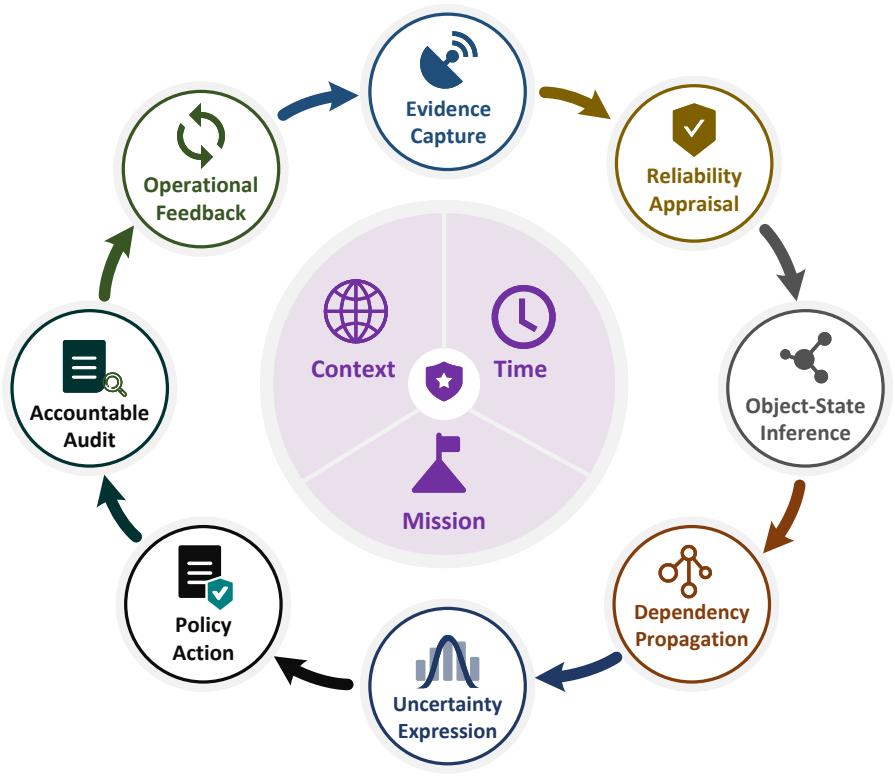


Figure 5. CTG appraisal and action loop, with policy, temporal state, and dependency knowledge shared across its steps.

6.4. Partition, Revocation, and Reconciliation Trace

The coalition in Section 3.1 has jointly approved policy version 1 and a low-consequence local delegation expiring at $t = 480$ s. At $t = 0$, the USV loses its shore connection; at $t = 240$ s, shore organizations commit a revocation of a UUV signing key, effective immediately. The UUV receives the revocation at reconnection, $t = 600$ s. Its protected clock reports event intervals, not exact event times. This scenario assumes a trustworthy pre-compromise time/counter record for the retained events; a stolen key alone cannot establish a credible historical timestamp.

Three signed records have intervals $[114, 126]$, $[234, 246]$, and $[294, 306]$ s. On reconnection, a verifier checks their source signatures, batch inclusion, policy epoch, and the revocation history. A record whose interval ends before 240 s can be eligible for the historical audit. One beginning at or after 240 s is revoked at event time. An interval crossing 240 s is quarantined for review. Historical eligibility does not authorize a new physical action or refresh the observation.

The illustrative mean averages signature validity, observation quality, lineage, cached policy match, and temporal authority, using $[1, 0.95, 1, 1, a]$ with $a = 1, 0.5, 0$ and threshold 0.75. Its compensating aggregation accepts all three records. A scalar policy with an explicit authority veto produces exactly the typed rule's decisions. The decision difference follows from whether the policy enforces temporal authority as a veto. Table 8 reports the replay outcomes.

Table 8. Executable replay outcomes. Ledger and policy-validating witnessed-log paths use identical evidence and authorization rules. The scalar examples use the same input features.

Event Interval (s)	OAL-Organized Rule / Validating Log	Mean Score	Scalar Decisions
114–126	Eligible historical record	0.99	Mean accepts; veto-aware scalar agrees with rule
234–246	Quarantine: revocation overlap	0.89	Mean accepts; veto-aware scalar quarantines
294–306	Revoked at event time	0.79	Mean accepts; veto-aware scalar rejects

During disconnection, an action at 420 s can have restricted local authority under the cached envelope even though a later audit identifies a conflict with effective global revocation; at 540 s the envelope has expired and the local policy denies the action. Reconciliation retains both the locally available authority and the later global judgment. Rewriting the old decision as if the vehicle had received the revocation at 240 s would erase this distinction.

The synthetic replay verifies Ed25519 signatures and Merkle inclusion, rejects altered records/proofs, checks all six arrival orders with duplicate delivery, and enumerates all 16 coalition signer subsets. The ledger-policy and validating-log decision functions agree throughout. Section 7.3 specifies how the same application rules map to Fabric and how the constrained vehicle receives an auditable result.

7. Blockchain-Assisted Assurance: Selection and a Specified Pattern

7.1. The Decision to Use a Ledger

Ledger adoption begins with writers, state, and authority. Wüst and Gervais organize the decision around the need to store shared state, the number and trust relationships of writers, and the availability of a trusted party [91]. Locher et al. examine which functions of a trusted third party can be replaced by distributed agreement [92]. For mission assurance, the relevant requirement is that no single organization may silently change accepted membership, policy, or revocation history, while authorized parties retain a mutually verifiable state.

In the coalition example, A alone must not replace a sensor-acceptance policy, C must not remove a previously acknowledged batch after a gateway incident, and D must not declare its own data accepted by the coalition. Four independent organizations can jointly validate those transitions. A permissioned ledger is one implementation of this requirement.

The strongest alternative is a transparency service whose checkpoints and state transitions are independently witnessed and checked under the same authorization rules. Such a service can satisfy the same safety requirements. A consistency-only witness detects or exposes inconsistent histories but does not by that act establish that a policy change was authorized. Adding joint policy validation closes that gap. The replay in Section 6.4 consequently finds equivalent decisions for the ledger and policy-validating log. Architecture selection then turns on implementation complexity, privacy, availability, recovery, and measured cost. Figure 6 summarizes this shared-state selection.

Governance viability is also an adoption condition. Maersk and IBM discontinued TradeLens after it failed to achieve the required global industry collaboration and com-

mercial viability, despite a technically viable platform [93]. This maritime case motivates evaluating incentives, participation, exit, and operating responsibility.

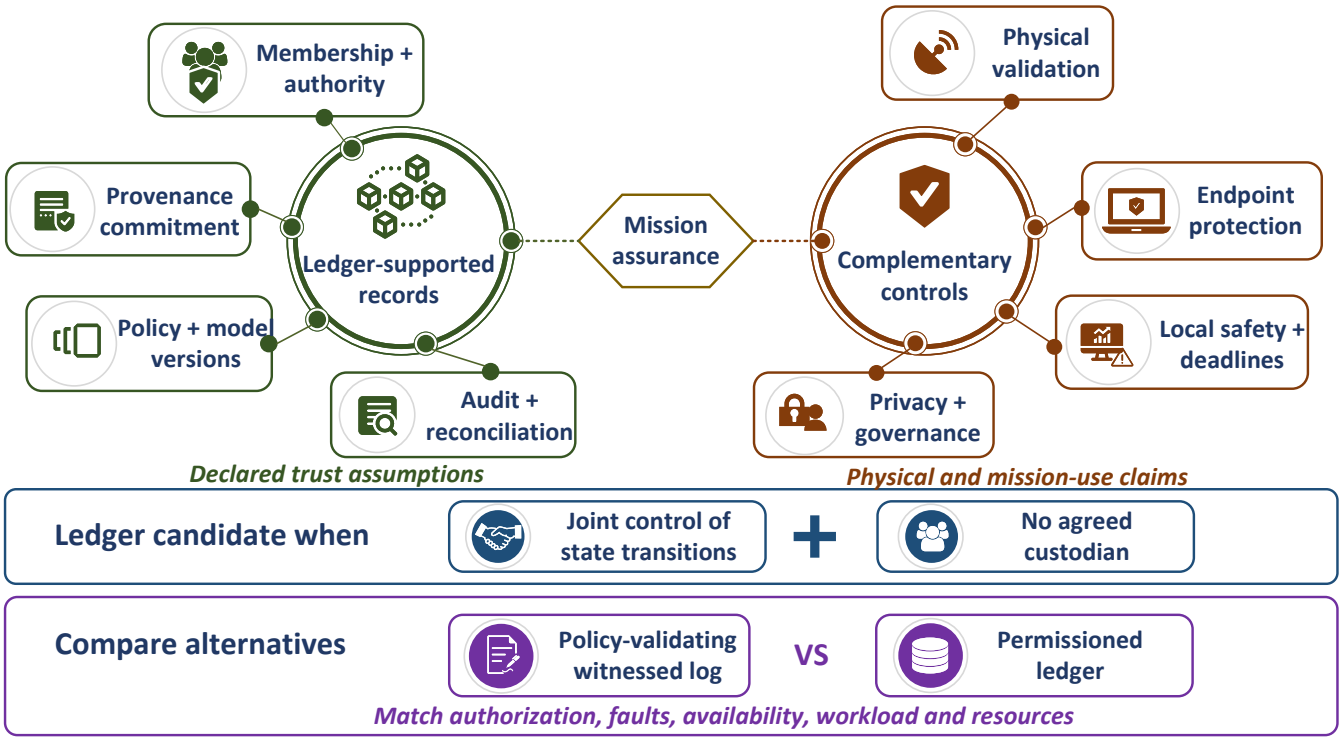


Figure 6. Requirement-based shared-state selection. A policy-validating witnessed log and a permissioned ledger are compared under matched governance guarantees. Physical appraisal and local safety remain application responsibilities.

7.2. Mission Requirements and Consensus Candidates

A requirement-based selection process, as advocated by Švarcmajer et al. [94], specifies the fault model before choosing a protocol. Here the coalition design assumes at most one Byzantine administrative domain among four independently controlled shore nodes. Safety must hold during a USV partition; new global decisions may stop when a responsive quorum is unavailable. The illustrative reconciliation deadline is 60 s after an event becomes available for shore processing. Membership is stable during a mission and changes through a jointly approved epoch transition. Shore nodes have adequate compute and power; UUVs do not participate in consensus. Consensus traffic uses shore/radio infrastructure, with only compact evidence commitments crossing the acoustic link.

Table 9 compares representative families. Normal-path message complexity is a selection indicator, conditional on batching, signature aggregation, dissemination, and view changes. Fault tolerance and liveness assumptions remain decisive. In particular, a DAG organization does not remove the need to state which agreement protocol orders or certifies its data.

Raft and Byzantine protocols solve different fault problems [81,83,84]. CometBFT uses its own round-based consensus and gossip rules [95]; Narwhal is a dissemination/mempool component, while Tusk supplies an associated ordering protocol [82]. Authority-based UAV acquisition, as implemented in BUS, illustrates placement on capable validators [16]. Lightweight IoT proposals similarly motivate moving consensus away from constrained endpoints [96]; the optimization must be assessed together with the resulting administrative dependencies.

Table 9. Consensus candidates against the mission requirements. Message orders describe representative normal paths; recovery can cost more.

Family	Fault Model and Partition Behavior	Latency, Traffic and Resource Implications	Scenario Fit
Raft	Crash faults; majority needed for progress	Leader replication, typically $O(n)$ exchanges per entry; low server cost; managed membership	Single accountable operator with crash-only replicas; insufficient for Byzantine coalition administrators
Classical BFT / SmartBFT	Byzantine faults; typically $n \geq 3f + 1$ and $2f + 1$ responsive votes	Multi-phase communication, commonly $O(n^2)$; timeout/view-change cost; capable stable nodes	Small independent shore coalition; selected Fabric ordering candidate
HotStuff	Byzantine faults; quorum and partial-synchrony liveness	Linear normal-path communication with quorum certificates and pipelining; cryptographic aggregation and leader availability matter	Larger capable coalition with measured load benefit
CometBFT	Byzantine faults; more than two-thirds voting power; round/locking rules	Tendermint-style propose/vote and gossip; timeouts under delay; implementation-dependent dissemination cost	Application-specific replicated state machine; not classified as HotStuff merely because both use BFT
Authority-based	Named validators; resilience depends on the actual protocol	Low mining cost; schedule/finality and quorum assumptions must be identified	Controlled consortium only after the concrete authority protocol meets the fault requirement
DAG-based (Narwhal/Tusk)	Reliable data dissemination plus an explicit BFT ordering protocol	Parallel dissemination and storage; asynchronous progress assumptions and communication cost remain	High-throughput shore service with resource evidence; no acoustic-partition shortcut
IoT-oriented designs	Often cluster heads, trusted managers, or restricted validator sets	Reduce endpoint work by moving it to capable nodes; altered trust and availability assumptions	Gateway-mediated collection; selected only if its delegated trust matches the mission

Fault assumptions are stated separately for infrastructure and applications. A Byzantine ordering service concerns agreement despite arbitrary behavior of replicas; it does not prevent an authorized producer from supplying a false observation. Endorsement constrains who approves a proposed state update, while the chaincode defines the deterministic rule being approved. A contract can consistently implement an unsuitable rule, so policy approval also requires evidence about its mission consequence. The application must specify whether emergency authority can bypass ordinary endorsement and how that exception is recorded and reviewed. Without an explicit exception, loss of a mandatory signer is an intentional loss of fresh global authorization.

Partition tolerance likewise has two meanings that must be separated. Safety preserves a consistent accepted history when messages are delayed or unavailable. Availability concerns whether a particular request can obtain the needed ordering and application approvals before its deadline. A shore quorum may remain healthy while the USV is disconnected, allowing shore revocations to progress while a UUV continues under cached authority. If the shore quorum itself is divided, new global updates may stop even after the gateway returns. Evaluations should vary these two partition locations independently. Classical consistency/availability results motivate this distinction under network partition [97]; they do not specify the vehicle's local safety policy.

Three scenario decisions follow. A single-agency inspection mission can use an authenticated database or witnessed log, adding Raft replication for crash availability. The four-organization response mission requires Byzantine ordering plus joint application

authorization, or an equivalent validating log; current Fabric SmartBFT is a concrete candidate [98]. A disconnected UUV group uses signed local records, store-and-forward transport, and bounded delegation, with no requirement for acoustic consensus. Increasing validator count only helps when it increases independent fault domains while preserving a reachable quorum. Actual latency and energy must be measured for the selected deployment.

7.3. A Coalition Provenance-And-Revocation Pattern

The specified pattern combines evidence commitments and post-partition reconciliation; Table 10 relates these to membership, version coordination, and audit.

Table 10. Ledger roles and their corresponding application requirements.

Role	Shared State	Required Complementary Evidence
Membership	Role, delegation, key epoch	Identity proof and protected key custody
Commitments	Batch root, sequence, retained-object reference	Signed source records and retrievable payload
Version coordination	Policy/model hash and approval epoch	Model validation and deployment evidence
Audit	Decision, evidence links, and effective-time history	Complete capture and independent review
Reconciliation	Received batch, gaps, conflicts, judgments	Protected clock/counter and offline-authority history

Placement and endorsement. Each organization A–D hosts an endorsing peer and a separate ordering node on capable shore infrastructure. The illustrative BFT service has four orderers, tolerating one Byzantine node under its protocol assumptions. Administrative separation supplies the independent fault domains. The application endorsement rule is `AND(A, B, OR(C, D))`; channel configuration and chaincode-definition approval require the same coalition authority. A and B are mandatory, so application progress can stop even while three orderers remain responsive. The USV forwards and caches signed evidence, with no consensus vote. A partitioned gateway buffers batches and retries idempotently after reconnection. UUV safety control uses its local envelope throughout.

State and encoding. The state keys are mission/policy/epoch, producer/key/history, delegation/identifier, batch/producer/sequence, and judgment/record/policy. The fixed-width source body contains one byte each for version and type, two 16-byte mission/producer identifiers, four-byte sequence and authority-epoch fields, two eight-byte event-time bounds, and three 32-byte hashes for policy, payload, and parent/root. These fields total 154 bytes. Integers use network byte order; time is signed seconds in the declared mission epoch. A 64-byte Ed25519 signature gives a 218-byte record. Domain-separated signing and hashing bind record type and version. A batch uses the same body layout with a Merkle root and previous-batch root, plus a two-byte record count: 156 bytes before signature and 220 bytes signed. Ten signed records are retained off chain. The gateway's canonical JSON application payload is 525 bytes.

Deterministic chaincode operations. `CommitBatch` reads the mission epoch, gateway role, and last batch sequence; verifies the compact gateway signature and policy binding; rejects an inconsistent sequence or duplicate identifier with different content; and records the root. An identical retry returns the existing result. This commits an auditable batch, not a claim that every leaf is fit for use. `JudgeRecord` verifies a supplied source signature and inclusion proof, reads the producer's historical key status and delegation, and writes a

separate eligible/revoked/quarantined judgment using the interval rule in Section 6.4. `UpdatePolicy` and `RevokeKey` require the coalition endorsement rule and monotone epochs; revocation records preserve their effective time. All state that is read for a decision enters the Fabric read set, allowing validation to reject stale concurrent updates. Chaincode obtains no external clock or oracle value during execution; authorized submitted evidence supplies those inputs.

Commit status and constrained verification. Fabric ordering precedes peer validation, so block inclusion is not proof of a valid world-state update [62,99]. After observing a valid commit, three distinct organizations satisfying A+B+(C or D) sign an application receipt containing the mission, policy/authority epoch, batch root, sequence, event span, and count. Receipt-signing code must check the commit validation status and the expected application state. A UUV checks authorized signer identities against its pinned epoch, signatures, monotone checkpoint sequence, source-record signature, and the Merkle branch. For ten leaves padded to 16, the branch has four 32-byte hashes. These checks establish a jointly attested application checkpoint and inclusion under pinned keys; they do not constitute native verification of all Fabric consensus and peer-validation execution. A new epoch requires an authenticated transition from the pinned epoch. The receipt is an additional application trust boundary that must be implemented and tested.

7.4. Communication Budget and Partition Sensitivity

The receipt and retrieval paths serve different evidence needs. A vehicle receiving a checkpoint can verify that independent organizations acknowledged the committed root under a pinned authority epoch. To appraise an observation, it still obtains the relevant leaf, its inclusion path, and any required source or transformation evidence. A commitment-only contact therefore supports a narrower claim than complete evidence delivery. Retrieval policy can prioritize recent high-consequence products and defer bulk imagery until a higher-bandwidth contact, while recording which required inputs remain unavailable. A failed retrieval is represented as missing evidence rather than a successful integrity check.

Clock and key transitions also affect the usable lifetime of a proof. A cached checkpoint remains evidence about a historical commitment after a new policy is issued, but its authority to support a new decision depends on the relying policy and authenticated transition history. A vehicle with an uncertain clock cannot establish that an expiry condition still holds by inspecting a ledger root. It needs a bounded time interval or a more restrictive fallback. Similarly, a fresh receipt cannot make an untrustworthy acquisition timestamp credible. These requirements bind the application-level proof to the same temporal assumptions used in the replay and define what a constrained verifier must do when those assumptions fail.

Ed25519 signatures contain 64 bytes [100]; ML-DSA-44 signatures contain 2420 bytes [101]. The budget compares these two compact application signature profiles. Fabric enrollment certificates, endorsements, proposal responses, read/write sets, and envelopes add deployment-dependent bytes. Only the gateway transports those full transactions on the shore network. Full transaction sizes and resource costs depend on the configured Fabric implementation; the 525-byte JSON value covers the gateway's application payload.

For a message of B bytes and net payload rate R bit/s, the serialization lower bound is $8B/R$. Table 11 uses 100, 1000, and 5000 bit/s as scenario inputs. It excludes propagation, contact waiting, modem framing, BPSec overhead, loss, retries, and certificate/epoch updates. Those factors increase end-to-end delay and must be added for a specified modem and transport configuration.

Table 11. Application-message serialization lower bounds. All sizes are computed from the encoding in Section 7.3; ML-DSA values substitute the standardized signature length.

Message	Bytes	100 bit/s	1 kbit/s	5 kbit/s
Ed25519 source record	218	17.44 s	1.744 s	0.349 s
Ed25519 batch commitment (10 records)	220	17.60 s	1.760 s	0.352 s
Ten complete Ed25519 records	2180	174.40 s	17.440 s	3.488 s
ML-DSA-44 batch commitment	2576	206.08 s	20.608 s	4.122 s
Ed25519 record + branch + joint receipt	697	55.76 s	5.576 s	1.115 s

The 697-byte downlink comprises a 218-byte source record, 128-byte branch, two-byte leaf index, 156-byte checkpoint, one-byte signer bitmap, and three 64-byte signatures. At one ten-record batch per minute, the 220-byte commitment needs about 29.3 net bit/s on average, whereas sending all ten signed records needs about 290.7 bit/s. Commitment-only delivery therefore fits some contacts that cannot carry full evidence, at the cost of later retrieval and incomplete immediate appraisal. Sending one ten-record ML-DSA-44 batch commitment at the same cadence requires about 343.5 bit/s. Public keys are assumed provisioned; key distribution and post-quantum verification time remain separate costs.

The partition sensitivity calculation covers 63 configurations: validator counts of 4, 7 and 10; responsive counts of quorum minus one, quorum, and all validators; and partition durations of 0, 10, 30, 58, 59, 60, and 120 s. With assumed post-contact delays of 2 s for a ledger path and 1 s for a validating-log path, a 60 s deadline permits partitions of at most 58 and 59 s, respectively, when a quorum is reachable. Below quorum, neither completes a fresh joint decision. Application endorsement availability must be checked in addition to ordering quorum.

8. Evaluation Practices and Evidence Maturity

8.1. A Claim-Specific Evidence Profile

L0–L5 record how a claim is supported: concept, analysis, simulation/data, prototype, field trial, and sustained operation. Figure 7 relates each evidence form to its inputs, reasoning or evaluation, and outputs. Formal soundness, experimental control, environmental representativeness, and operational duration remain separate assessment dimensions.

Within the 39-study sample, six profiles include L0, eleven L1, twenty L2, twenty L3, and one L4; none includes L5. Profiles overlap; four studies are conceptual-only. The L4 assignment concerns S31's bounded physical UAV-node cooperation/routing experiment; its container consensus benchmark remains L3. Treating that experiment as L3 only is evaluated as a sensitivity variant in Section 5.4. BUS supplies hardware-prototype evidence, while recorded GNSS trajectories support L2. These settings answer different questions about implementation, physical operation, and transfer.

As the comparisons in Section 5.3 show, evidence form, parameter strength, exercised mechanism, and environmental relevance jointly determine the supported claim.

8.2. Datasets, Simulators, and Testbeds

Public resources support complementary evaluation tasks. CICIoT2023, TON_IoT, and Edge-IIoTset supply attack and telemetry data for classification and robustness tests [102–104]. SeaDronesSee supplies maritime visual data [105]. AirSim, UUV Simulator, and Aqua-Sim NG address vehicle/sensor simulation, underwater dynamics, and underwater communication, respectively [106–108]. Their use in one experiment requires compatible clocks, geometry, mobility, and event interfaces, together with a separately validated mission-loss model. Table 12 summarizes the resources and transfer checks.

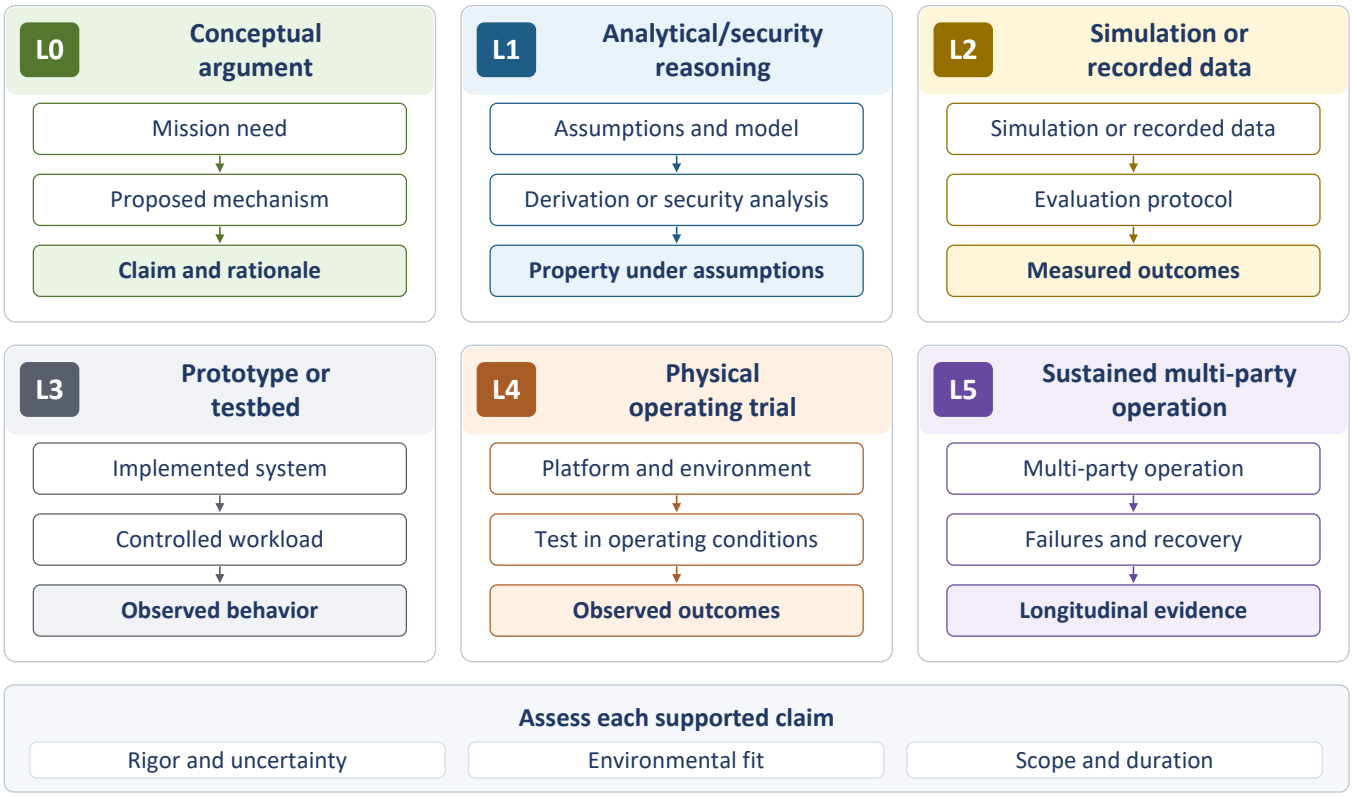


Figure 7. Complementary evidence forms and their claim-specific support. Each panel relates inputs, reasoning or evaluation, and outputs; rigor, environmental fit, and scope are assessed separately.

Table 12. Evaluation resources and required transfer checks.

Resource	Useful Evidence	Required Complement
IoT attack/telemetry datasets	Detection, calibration, missingness and false isolation	Maritime topology, benign degradation and mission consequences
SeaDronesSee / GNSS recordings	Visual detection or signal appraisal	Independent ground truth for the stated physical claim
AirSim / UUV Simulator	Dynamics, sensors, disturbance and control	Field validity of the configured models
Aqua-Sim NG	Path delay, loss, energy and contact behavior	Physical observation and governance modules
Ledger and witnessed-log testbeds	Validation, bytes, latency, recovery and membership changes	Matched guarantees, evidence workload and fault domains
Multi-organization field exercise	Integrated decisions, recovery and audit	Repeated scenarios and explicit operational duration

8.3. Metrics, Baselines, and Reproducibility

An evaluation should measure inference error and uncertainty, mission consequences, network/resource cost, shared-state behavior, and accountability. Mission outcomes include unsafe acceptance, false isolation of a critical relay, task completion, safety-constraint violations, and deadline misses. Resource measures include bytes per evidence event, energy, and endorsement/commit latency. Accountability measures include the time to

identify affected descendants and the fraction of decisions with retrievable provenance. Reporting only throughput omits the consequence of a timely but unsuitable decision.

Scenario factors include compromised entities versus devices, independent versus correlated observations, fading versus jamming, fresh versus stale attestation, lost contacts, collusion, and different authority-expiry periods. Ground truth distinguishes benign degradation, adversarial intervention, and their coexistence; unknown is an appraisal outcome. Training, tuning, calibration, and evaluation use separate data, with splits respecting shared missions, platforms, and recording episodes. Paired scenarios and uncertainty intervals over independent runs support comparative claims. Undelivered events and deadline misses remain in the latency/availability analysis.

Baselines receive the same raw evidence, action set, mission objectives, and tuning budget. Appraisal comparisons include a scalar model with explicit authorization vetoes, and component ablations for lineage, shared causes, and uncertainty. Storage/governance comparisons include an authenticated database, an independently witnessed consistency log, and a policy-validating jointly cosigned log. The strongest baseline matches administrative independence, quorum, retention, and effective-time semantics. Differences in guarantees are reported before interpreting differences in cost.

Reproduction requires the topology, fault domains, and channel/contact models. The implementation description should give software and policy/model versions, payload encodings, cryptographic suites, and endorsement rules. Workload, seeds, ground truth, and mission-loss weights define the evaluation conditions.

Evaluation should also distinguish collection completeness from record consistency. A test can maintain an independent event oracle, then inject omission before signing, deletion of an unwitnessed suffix, relay withholding after commitment, and loss of an off-chain object after its root has been accepted. These failures have different observability. A retained receipt exposes the absence of an acknowledged batch; it cannot reveal an event never captured by any trusted observer. Metrics should therefore report the denominator of events expected to be captured, those acknowledged by a witness, those committed under joint policy, and those whose payloads remain retrievable. This separates preservation, detection, and recovery rather than treating all three as one audit-success rate.

Recovery workloads need similar precision. A returned gateway may carry valid old observations, revoked-key submissions, conflicting sequence numbers, and records generated under different policy versions. Benchmarking only clean insertion misses this mix. Experiments should vary backlog age and size, compromised-key cutoff, duplicate rate, epoch transitions, proof retrieval, and the proportion requiring human review. Report both automated processing time and the time until the application can act on a resolved judgment. Shared-state throughput can remain high while semantic reconciliation is blocked by missing evidence or unavailable mandatory endorsers.

Finally, comparison of inference policies should preserve the cost of their actions. Quarantining every uncertain observation can reduce unsafe acceptance while preventing useful missions. A declared loss model assigns costs to unsafe reliance, unnecessary isolation, deferred tasks, corroboration requests, and human review. Results should show sensitivity to those weights and to the prevalence of attacks and benign degradation. Calibration and decision loss answer different questions: a well-calibrated estimate may still be used by an unsuitable policy, while a conservative rule can make safe decisions without producing probabilities. Keeping the two evaluations separate follows the appraisal/decision interface in Equation (1).

9. Discussion and Research Agenda

9.1. Integration, Governance, and Privacy

Composition requires both claim-specific policy and dependency analysis. Architectural common causes remain consequential: adding UUVs does not remove their dependence on one gateway, clock, or operator.

Consortium design must specify admission, contract/model approval, emergency authority, disputes, member exit, and audit responsibility. Maritime adoption research and the TradeLens experience show why participation and incentives belong in this analysis [20,93,109]. The endorsement rule in the example preserves joint control while creating a measurable availability cost through its mandatory signers. Additional replicas under one owner do not create additional independent governance domains.

Cross-sector studies distinguish traceability benefits from the organizational changes required to realize them [110,111]. Coalition agreements assign retention, disclosure, maintenance, and incident-response duties, including evidence availability after member exit and historical signatures after key rotation. Ledger and validating-log comparisons must match these governance provisions.

Privacy depends on more than payload encryption. Stable identifiers, route custody, endorsement timing, and model-use records can reveal vehicle activity and organizational relationships [112,113]. Data minimization, selective disclosure, retention, and controlled identity resolution therefore accompany auditability. A revocation invalidates future reliance under a policy; it does not erase a retained commitment. Human override is a separately authorized, recorded transition with its evidence, rationale, and consequence.

9.2. Falsifiable Research Propositions

The following propositions define future comparative tests. Each experiment prespecifies its mission distribution, minimum useful effect, primary outcome, and analysis rule. Null and adverse results identify conditions where the additional structure provides no benefit.

P1 (claim organization). Test whether an assurance vector reduces held-out unsafe acceptance at matched false-isolation and latency budgets against a tuned scalar policy. Both receive complete evidence, identical actions, and equal tuning budgets; the scalar includes authorization/freshness gates and interactions. Equal or better comparator performance refutes the proposition, as the veto-aware scalar illustrates in Table 8.

P2 (dependency modeling). Under prespecified shared-source error and collusion rates, test whether a model with common-cause variables improves a proper scoring rule and mission loss relative to an equally tuned independence model. Include correct, incomplete, and erroneous lineage conditions. The proposition fails when model misspecification or acquisition cost offsets the gain, or when observations are sufficiently independent that no useful improvement remains.

P3 (diagnosis and unknown status). Compare a model retaining benign, adversarial, and unresolved hypotheses with a binary appraisal model under mixed degradation, attacks, and evidence outages. Both can choose the same actions, including abstention. Separate diagnosis and abstention ablations, and measure loss from unsafe acceptance, false isolation, corroboration delay, and lost availability. The proposition fails if the added diagnostic state increases loss or yields no useful improvement.

P4 (offline authority). Vary delegation scope, expiry, clock uncertainty, partition duration, and revocation time. Test whether bounded delegation offers a useful trade-off between tasks completed before deadline and actions conflicting with effective global policy, compared with fresh-global-only authorization and unrestricted offline autonomy.

Report locally authorized actions and retrospectively conflicting actions separately. Failure includes operating regimes where the bounded policy is dominated in both outcomes.

P5 (conditional ledger value). Compare implemented ledger and independently witnessed-log services under matched membership, faults, rules, retention, and workload. Vary n, responsive quorum, partition duration D, event rate, and backlog. For deadline τ, estimate $D + L(n, \lambda) \leq \tau$ using measured post-contact delay L at load λ. Test improvement in timely verifiable decisions or recovery completeness at fixed byte/energy budgets. Equal safety with greater cost, lower endorser availability, or no improvement over the log refutes the advantage in that regime. Section 7.4's 58/59 s boundary uses assumed delays.

P6 (evaluation coverage). Select configurations from identical candidate sets and equal evaluation budgets using either transaction/network metrics alone or those metrics plus observation, endpoint, governance, and mission outcomes. Compare held-out mission loss under fixed selection rules. The proposition fails if the broader evaluation does not improve selection after its additional acquisition and interpretation costs are included.

9.3. Assurance Cases and Maritime Requirements

OAL supplies the content of an assurance argument; existing notations supply its structure. GSN represents claims, strategies, context, assumptions, and evidence, while SACM provides a structured assurance-case metamodel [114,115]. UL 4600 supplies safety-case requirements for autonomous products rather than a diagram notation [116]. OAL can annotate their claims with object/version, assurance layer, lifecycle stage, event interval, and evidence profile. Conformance assessment remains tied to the requirements of the applicable standard.

Route-replanning suitability requires credible observations, protected derivation, valid authority, current policy, and acceptable uncertainty. The argument records observation, issue, receipt, and expiry times, including cached authority. Corrected inputs or revoked keys trigger descendant review while preserving independently supported products. Validation should measure evidence gaps, detection of seeded invalid arguments, reviewer error, and time to identify affected products.

Maritime requirements add a specific application boundary. IMO's interim guidelines for MASS trials and regulatory scoping exercise provide trial and regulatory context [117,118]. The non-mandatory MASS Code was adopted in May 2026 and took effect on 1 July 2026 [119]. DNV-CG-0264 addresses autonomous and remotely operated ships [120]. An assurance case must identify the vessel and operation to which these instruments apply, the responsible authority, operational design conditions, human/remote-control responsibilities, and evidence of safe fallback. Their maritime scope cannot be transferred wholesale to UAV or underwater operations. A ledger record can trace an approval or operating limit; the substantive safety evidence and applicable approval process remain separate.

9.4. Toward a Cross-Domain Unmanned Trust Profile

The proposed Cross-Domain Unmanned Trust Profile (CDUTP) is an interoperability agenda for versioned claims and evidence references. It can reuse RATS roles, W3C PROV relations, zero-trust policy concepts, and secure-update manifests [24,26,28,121]. UAS and maritime service specifications supply communications context [122,123]. A profile instance records object/type, scoped claim, issuer and relying role, evidence reference, event and receipt times, expiry, uncertainty, permitted use, policy/model/schema versions, disclosure conditions, dependencies, and revocation binding.

The compact encoding in Section 7.3 is one illustrative transport profile. Interoperability requires version-specific semantic mappings and tests, including bounded clocks,

key rotation, missing evidence, and disconnected issuance. Independent implementations can then be compared in plugfests and multi-organization exercises. The key research question is whether common semantics reduce decision and audit errors at an acceptable communication and governance cost.

10. Limitations

The 39-study analytical sample is purposively selected from studies with inspectable full texts and evaluations. Coverage depends on the search and access conditions, and the extraction has no independent inter-rater estimate. Sensitivity analyses characterize changes caused by domain balance, conceptual studies, and selected coding choices; they cannot measure missing-literature bias. In particular, the dominant cell differs between the direct-domain and transferable strata. The single L4-coded study supports a bounded UAV-node experiment, while L5 and integrated maritime validation remain unobserved in the extracted claims. Neither these absences nor empty heatmap cells establish field-wide gaps.

OAL and CTG organize claims, dependencies, and policy interfaces. Application-specific validation is needed for inference rules, calibration, dependency completeness, correlation estimates, causal interpretations, model transfer, and evidence sufficiency. A scalar policy can preserve the same constraints as a structured representation, as the worked case demonstrates. Claims of improved mission outcomes therefore require the matched comparisons in Section 9.2.

The worked case uses synthetic events, a fixed coalition, protected-clock assumptions, and declared revocation semantics. It checks signatures, inclusion, deterministic policy rules, and byte lengths. It does not execute Fabric, measure consensus or acoustic performance, implement a production receipt service, or validate integrated UAV–USV–UUV safety. Deployment measurements are needed for full Fabric transaction sizes, provisioning overhead, and cryptographic energy. Operational tests must cover clock rollback resistance, modem behavior, receipt issuance, and operator response. Communication times are serialization lower bounds; quorum sensitivities use assumed delays.

Cryptographic evidence depends on protected keys, credible capture, retrievable payloads, and authenticated time or counters. An attacker who controls an unprotected key and clock can fabricate apparently historical records. A Merkle root does not recover withheld data or prove that an event was captured. Joint validation depends on genuinely independent administrative domains and an available application endorsement set. Governance deadlock, collusion beyond the fault model, and disputed physical facts require procedures outside consensus.

Transfer to maritime operations requires environmental and regulatory assessment for the actual vehicle, mission, and jurisdiction. Public datasets and component prototypes do not determine the consequences of rare faults in a deployed coalition. Field studies should evaluate privacy, retained metadata, human authority, recovery, and sustained organizational participation alongside technical behavior.

11. Conclusions

Mission trust connects evidence about physical observations, protected records, jointly governed state, and fitness for a particular use. OAL organizes these claims by object and lifecycle, while a typed dependency graph and CTG loop identify when changes require reassessment. Applying the framework to 39 full-text studies distinguishes complementary mechanisms, their evaluated settings, and the evidence needed at each handover. Stratified and sensitivity analyses make the dependence of coverage patterns on selection and coding visible.

The partition–revocation case specifies coalition roles, a Fabric-oriented application pattern, constrained verification, and a reproducible communication budget. Its executable replay preserves the distinction between local authority, event-time revocation, and later audit. A policy-validating witnessed log produces the same decisions under matched rules. Ledger value is therefore conditional on shared-governance requirements and the measured cost of implementing them.

The resulting research agenda calls for independently reproducible coding, matched architecture and appraisal baselines, realistic partition and recovery tests, and integrated mission evidence. The framework provides a traceable basis for matching mechanism evidence to mission requirements and selecting the governance service that those requirements justify.

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Abbreviations

The following abbreviations are used in this manuscript:

AUV	Autonomous underwater vehicle
CDUTP	Cross-Domain Unmanned Trust Profile
CTG	Context–temporal–graph
DLT	Distributed ledger technology
OAL	Object–Assurance–Lifecycle
RATS	Remote ATtestation procedureS
UAS	Unmanned aerial system
UAV	Unmanned aerial vehicle
USV	Unmanned surface vehicle
UUV	Unmanned underwater vehicle
ZTA	Zero trust architecture

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